

CITY DIAGNOSTIC CENTRE

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COMPREHENSIVE BLOOD INVESTIGATION REPORT

Patient Name:	Avinash Prakash	Report No:	CDC/2026/BL-4821
Age / Gender:	19 Years / Male	Sample Date:	06-May-2026, 08:30 AM
Ref. Doctor:	Dr. Ramesh Iyer (Orthopaedics)	Report Date:	06-May-2026, 02:15 PM
Sample Type:	Venous Blood (EDTA + SST)	Barcode:	826-4821-0906

1. COMPLETE BLOOD COUNT (CBC)

Investigation	Result	Reference Range	Unit	Flag
Haemoglobin (Hb)	10.8	13.0 - 17.0	g/dL	LOW
RBC Count	3.62	4.5 - 5.5	x10 ⁶ /uL	LOW
Haematocrit (PCV)	32.4	40.0 - 50.0	%	LOW
MCV	84.2	80.0 - 100.0	fL	
MCH	28.6	27.0 - 33.0	pg	
MCHC	33.1	31.5 - 34.5	g/dL	
RDW-CV	14.8	11.5 - 14.5	%	HIGH
WBC (Total Leucocyte Count)	11,200	4,000 - 10,000	cells/uL	HIGH
Neutrophils	72	50 - 70	%	HIGH
Lymphocytes	20	20 - 40	%	
Monocytes	6	2 - 8	%	
Eosinophils	2	1 - 4	%	
Basophils	0	0 - 1	%	
Platelet Count	3,42,000	1,50,000 - 4,00,000	cells/uL	

2. BONE & TUMOR MARKER PANEL

Investigation	Result	Reference Range	Unit	Flag
Alkaline Phosphatase (ALP)	412	44 - 147	U/L	HIGH
Lactate Dehydrogenase (LDH)	587	140 - 280	U/L	HIGH
Serum Calcium (Total)	10.9	8.5 - 10.5	mg/dL	HIGH
Serum Phosphorus	5.1	2.5 - 4.5	mg/dL	HIGH
Serum Uric Acid	6.8	3.5 - 7.2	mg/dL	
C-Reactive Protein (CRP)	28.4	< 5.0	mg/L	HIGH
ESR (Erythrocyte Sedimentation Rate)	52	0 - 20	mm/hr	HIGH

* Elevated ALP and LDH are non-specific markers and may be associated with rapid bone turnover, hepatic pathology, or active bone lesions. Clinical correlation with imaging findings is strongly advised. These values alone are not diagnostic of malignancy.

3. LIVER FUNCTION TEST (LFT)

Investigation	Result	Reference Range	Unit	Flag
Total Bilirubin	0.8	0.2 - 1.2	mg/dL	
Direct Bilirubin	0.3	0.0 - 0.4	mg/dL	
SGOT / AST	34	10 - 40	U/L	

SGPT / ALT	28	7 - 56	U/L	
GGT	22	9 - 48	U/L	
Total Protein	7.1	6.0 - 8.3	g/dL	
Albumin	4.0	3.5 - 5.0	g/dL	
Globulin	3.1	2.0 - 3.5	g/dL	

4. KIDNEY FUNCTION TEST (KFT)

Investigation	Result	Reference Range	Unit	Flag
Blood Urea Nitrogen (BUN)	14	7 - 20	mg/dL	
Serum Creatinine	0.9	0.7 - 1.3	mg/dL	
eGFR	98	> 60	mL/min/1.73m2	
Serum Sodium (Na+)	138	136 - 145	mEq/L	
Serum Potassium (K+)	4.1	3.5 - 5.1	mEq/L	
Serum Chloride (Cl-)	101	98 - 107	mEq/L	

5. COAGULATION PROFILE

Investigation	Result	Reference Range	Unit	Flag
Prothrombin Time (PT)	12.4	11.0 - 13.5	seconds	
INR	1.02	0.8 - 1.2	ratio	
aPTT	30.2	25.0 - 35.0	seconds	

CRITICAL VALUES SUMMARY

The following values are significantly outside normal reference ranges and require immediate clinical correlation:

- 1. Alkaline Phosphatase (ALP): 412 U/L** — Markedly elevated (Normal: 44-147 U/L). Significant elevation suggests active bone remodeling or bone tissue pathology.
- 2. Lactate Dehydrogenase (LDH): 587 U/L** — Significantly elevated (Normal: 140-280 U/L). Elevated LDH is associated with tissue destruction and high cellular turnover.
- 3. ESR: 52 mm/hr** — Elevated, indicating active inflammatory or pathological process.
- 4. CRP: 28.4 mg/L** — Elevated, consistent with active tissue inflammation or destruction.
- 5. Haemoglobin: 10.8 g/dL** — Mild normocytic anaemia. May indicate chronic disease process.

RECOMMENDATION: These findings, particularly elevated ALP and LDH in a young male patient, warrant urgent correlation with clinical presentation and imaging studies.

Verified By:

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